

FIGMA: Efficient Instruction-Guided Adaptation of Multimodal Foundation Models via Automated Data Augmentation

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Abstract. Adapting multimodal foundation models under limited data and computational constraints remains a critical challenge. We propose FIGMA, an efficient instruction-guided adaptation framework for diffusion-based multimodal models via automated data augmentation. Starting from a small set of seed image-instruction pairs, our pipeline expands both visual and textual data using CLIP-based semantic extraction and diffusion-based image augmentation. We then apply lightweight LoRA fine-tuning, updating less than 1% of parameters, to adapt a pretrained Stable Diffusion model to domain-specific tasks.

Experiments show a 2.59 reduction in FID (60.51% over zero-shot) and a 19.23% improvement in CLIP Score, with strong cross-domain generalization across Nike, Zara, and FashionGen datasets.

Our contributions include: (1) a scalable multimodal augmentation pipeline for efficient adaptation, (2) a parameter-efficient instruction tuning strategy for diffusion models, and (3) an empirical analysis of data scale versus adaptation performance. Our code is available on GitHub³ and the data can be accessed on Zenodo⁴.

Keywords: Foundation Models · Efficient Adaptation · Multimodal Learning · LoRA · Data Augmentation

1 Introduction

With recent advances in computing resources, generative AI models that utilize large-scale data have been rapidly evolving. Research on instruction tuning for multimodal LLMs has been actively carried out to improve their ability to understand and perform complex tasks [2]. For example, LLaVA [8] enables multimodal instruction execution, ChartInstruct [10] focuses on chart reasoning tasks,

* G. Jung and M. Hong contributed equally to this work.

³ <https://github.com/Gayeon6423/FIGMA>

⁴ <https://doi.org/10.5281/zenodo.15628618>

and InstructCV [1] aligns diffusion models with instructions for computer vision tasks. However, adapting large multimodal foundation models to domain-specific tasks remains computationally expensive.

In the fashion domain, the ability to generate clothing tailored to specific styles, seasons, and purposes is essential. However, training such instructed fashion image generation models requires large-scale, high-quality image-instruction data. General image-text data often fail to adequately capture detailed clothing attributes.

This study proposes an automated data augmentation pipeline for image-text fashion data, utilizing CLIP [11] and Stable Diffusion [12]. The main contributions are: **(1) An automated pipeline for image-instruction data augmentation, (2) Enhancement of instruction-following capability through lightweight fine-tuning, and (3) Open-source release of the fashion image-instruction data.**

2 Related Work

2.1 Text-to-Image Generation Model

CLIP (Contrastive Language–Image Pre-training) learns joint representations of text and images by training on image-caption pairs, enabling zero-shot image recognition and retrieval [7]. Diffusion-based models such as Stable Diffusion have achieved high-fidelity image synthesis from text by iteratively transforming random noise into images [5,12].

2.2 Data Augmentation for Generative Model

Self-Instruct [16] is a semi-automated bootstrapping framework that leverages pretrained language models to create new instruction-following data. Applied to GPT-3 [9], it generated approximately 52,000 new instruction data points, resulting in a 33% improvement over the base model in the Super-Natural Instructions benchmark [17].

2.3 Instruction Fine-Tuning in Vision-Language Model

ChartInstruct introduces chart-specific instruction-following data with over 191,000 automatically generated instructions for more than 71,000 chart images [10]. InstructCV extends instruction tuning to computer vision tasks by reformulating segmentation, detection, and depth estimation as image generation problems guided by textual prompts [1].

2.4 Parameter-Efficient Adaptation of Foundation Models

Parameter-efficient fine-tuning methods such as LoRA [6] introduce low-rank adaptation modules to reduce the number of trainable parameters while preserving performance. Such approaches have been widely adopted for adapting large

language models and diffusion models under limited computational resources. Recent studies have explored lightweight and modular adaptation strategies to enable scalable deployment of large foundation models. Our work builds upon this line of research by combining automated data augmentation with parameter-efficient diffusion model tuning.

3 Method

The proposed method employs an automated pipeline that augments image data using an image-to-image model and augments instruction data by extracting and structuring key attributes. The overall architecture is illustrated in Fig. 1.

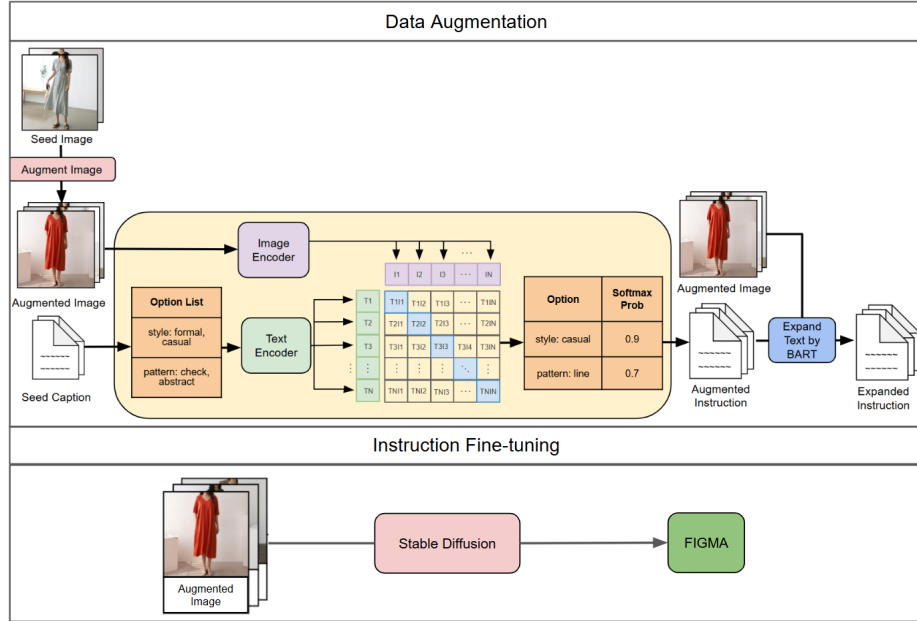


Fig. 1. (Top) The figure shows the process of image-instruction data augmentation using seed data; (Bottom) The figure below shows how the augmented data is used for instruction fine-tuning to build FIGMA.

3.1 An Automated Pipeline for Data Augmentation

Define Image-Instruction Data We randomly sample image-caption pairs from 10 out of 23 styles in the K-Fashion data⁵ and convert them into instruction format. We manually assign fashion-related attributes such as "appropriate

⁵ <https://aihub.or.kr/aihubdata/data/view.do?currMenu=115&topMenu=100&aihubDataSe=realml&dataSetSn=51>

occasion" and "suitable season" to each image. The final instruction data consists of: (1) **Prompt**: task description, (2) **Input**: captions with attributes, and (3) **Output**: image filename. We construct 230 seed image-instruction pairs.

Augment Image Data We leverage Stable Diffusion Img2Img Pipeline to augment the existing seed images. This process generates new images while maintaining key characteristics of the originals, significantly reducing the need for manual image collection and annotation. Our augmentation pipeline utilizes the "Realistic-Vision-V4.0-noVAE"⁶, selected for its capability in generating high-quality, realistic images. The prompt instructs the model to create a high-quality, full-body image of a person wearing a different outfit, while preserving the original image's style, composition and lighting. For each seed image, we generate five augmented versions using a strength parameter of 0.8 and guidance scale of 8.

Augment Instruction Data We leverage CLIP "clip-vit-large-patch14"⁷ to automatically generate instructions. The model was chosen because it effectively captures shared representations between images and text, enabling the automatic generation of contextually suitable text instructions for visual content. The pipeline consists of: (1) Prepare style option list extracted from seed captions (Style: Retro, Romantic, etc.; Color: Black, Red, etc.; Length: Regular, Long, etc.), (2) Extract options by computing cosine similarity between image and text embeddings, selecting the highest probability option, and (3) Generate instruction by combining selected options with predefined prompt and filename.

Algorithm 1 presents the instruction generation process. For each image, the CLIP model extracts image embeddings and compares them with text embeddings of predefined style options using cosine similarity. The most relevant options are selected via softmax, and structured captions are generated to form the final instructions. The proposed algorithm aims to generate augmented instruction data D' using a CLIP model M to associate augmented image data D with predefined style option list O . The process begins by initializing an empty data D' . For each image $I \in D$, an empty list L is created to store the selected style options. The algorithm then extracts the image embedding v_i using the CLIP image encoder.

To determine the most relevant style options, the algorithm iterates over each style option $o \in O$, encoding it into a text embedding v_o using the CLIP text encoder. The cosine similarity s_o between the image embedding v_i and each text embedding v_o is computed. After computing the similarity scores, the algorithm selects the most relevant style option and appends it to the list L . Based on the selected options, a structured caption C is constructed for the image I . Additionally, a predefined prompt P and a corresponding filename F are retrieved. These elements are then combined to form an instruction *Instruct*,

⁶ https://huggingface.co/SG161222/Realistic_Vision_V4.0_noVAE

⁷ <https://huggingface.co/openai/clip-vit-large-patch14>

which is stored in the augmented instruction data D' . This process is repeated for all images in D , and the final data D' is returned as the output.

Algorithm 1 CLIP-based Instruction Augmentation

Input: Augmented image D , Predefined style option list O

Output: Augmented instruction text D'

```

1: Load CLIP model  $M$ 
2:  $D' \leftarrow \emptyset$  {Initialize Augmented instruction text}
3: for each image  $I$  in  $D$  do
4:    $L \leftarrow \emptyset$  {Selected option list}
5:    $S \leftarrow \emptyset$  {Initialize similarity scores list}
6:    $v_i \leftarrow M.encode\_image(I)$  {Image embedding}
7:   for each option  $o$  in  $O$  do
8:      $v_o \leftarrow M.encode\_text(o)$  {Text embedding}
9:      $s_o \leftarrow cosine\_similarity(v_i, v_o)$ 
10:     $S \leftarrow S \cup \{s_o\}$ 
11:     $P_{opt} \leftarrow softmax(S)$  {Convert similarities to probabilities}
12:     $L_{selected} \leftarrow \arg \max(P_{opt})$  {Select the highest probability option}
13:     $L \leftarrow L \cup \{L_{selected}\}$ 
14:   end for
15:    $P \leftarrow predefined\_prompt$ 
16:    $C \leftarrow generate\_caption(L)$ 
17:    $F \leftarrow filename(I)$ 
18:    $Instruct \leftarrow combine(P, C, F)$ 
19:    $D' \leftarrow D' \cup Instruct$ 
20: end for
21: return  $D'$ 

```

Expand Instruction Data We refine augmented instructions using `bart-large-cnn`⁸, which provides strong abstractive generation capabilities. For each instruction, we generate 10 candidates using beam search and select the version with the highest CLIP similarity to the corresponding image, ensuring semantic consistency.

3.2 Instruction Fine-tuning for Text-to-Image Model

We fine-tune the pretrained Stable Diffusion model using LoRA [6], which adds low-rank adapters to specific UNet modules [13]. This approach enables training of fewer than 1% of model parameters, reducing computational cost while maintaining performance.

⁸ <https://huggingface.co/facebook/bart-large-cnn>

4 Experiments

4.1 Data

We utilized K-Fashion data consisting of 1,200,000 image-caption pairs across 23 style labels. We constructed 230 seed image-caption pairs and created augmented datasets at various scales. Nike⁹, Zara¹⁰, and FashionGen¹¹ datasets were obtained from Kaggle and GitHub. Data statistics are in Table 1, and examples in Fig. 2.

Table 1. Data statistics

Data	Size
Seed	230
1x Augment	230
5x Augment	1,150
10x Augment	2,300
50x Augment	11,500
100x Augment	23,000
Nike/Zara/FashionGen	200 each

K-Fashion Data		Augmented Data	
	Coordinate : X:299.5, Y:21.5 Style: Retro Substyle: Oriental Top: Length-Normal, Category-T-Shirt, Material-Woven Bottom: Length-ankle, Category-Pants, Material-Woven		It should be in photo format, not a drawing. Style: Sporty Substyle: Genderless Outerwear: Length-Normal, Color-Silver, Category-Jumper, Details-Zipper, Sleeve Length-Long, Material- Woven, Print-Camouflage, Neckline-Hood, Fit-Loose Bottoms: Length-Mini, Color- Silver, Category-Pants, Material- Woven, Print-Camouflage, Fit-Loose
Image	Caption	Image	Instruction

Fig. 2. (Left) Image-caption pair example of K-Fashion data; (Right) Image-instruction pair example of augmented data

4.2 Evaluation Metrics

We employed three evaluation metrics: **FID** [4] quantifies distributional difference between real and generated images (lower is better), **CLIP Score** [3] evaluates semantic coherence between text and images (higher is better), and **LPIPS** [18] measures perceptual similarity in feature space (lower is better).

⁹ <https://www.kaggle.com/datasets/thedevastator/nike-usa-products-prices-descriptions-and-custom>

¹⁰ <https://www.kaggle.com/datasets/xontoloyo/data-penjualan-zara>

¹¹ <https://github.com/menardai/FashionGenAttnGAN>

4.3 Human Evaluation

We conducted a user survey to evaluate the effectiveness of the augmented data and the generation quality of FIGMA. To assess whether the augmented instructions were appropriately generated for the corresponding images, we presented 10 images along with their respective instructions and asked 67 survey participants to respond according to the following evaluation criteria. The same criteria were also applied to assess whether the images generated by FIGMA accurately followed the given instructions. Participants evaluated 10 images using four criteria: Best (all attributes well-matched), Good (generally well-matched), Bad (poorly matched), and Worst (none matched). Results are shown in Fig. 3.

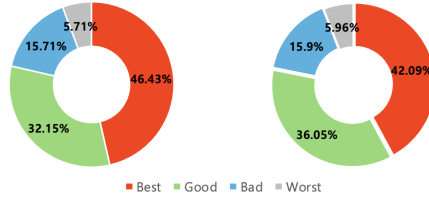


Fig. 3. (Left) Alignment rate for augmented data; (Right) Alignment rate for FIGMA-generated data

Participants also responded to each image with its instruction and selected all attributes they found applicable from the following five categories: *Style*, *Suitable occasion*, *Length*, *Color* and *Print*. Results are shown in Fig. 4.

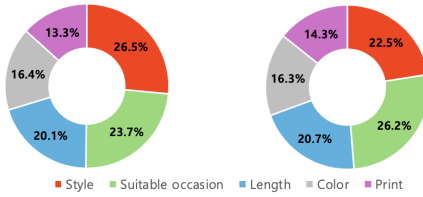


Fig. 4. (Left) Attribute match rate for augmented data; (Right) Attribute match rate for FIGMA-generated data

4.4 Training Setup

We fine-tuned the model using AdamW optimizer for 100 epochs on NVIDIA RTX 4090 GPU. We adopted LoRA [6] with rank 4, updating approximately 1%

of UNet [14] parameters. Training required a single RTX 4090 GPU, highlighting the computational efficiency of the proposed adaptation framework.

5 Experiment Results

5.1 Impact of Data Augmentation and Scale

The model trained with 1x augmented data showed a 1.84 reduction in FID compared to the zero-shot baseline. The 10x augmented data achieved the best performance with a 2.59 FID reduction (60.51% over zero-shot). In contrast, higher augmentation rates (50x, 100x) led to performance degradation, likely due to distributional drift introduced by excessive synthetic samples (Table 2). We refer to the 10x model as FIGMA.

Table 2. Impact of Data Augmentation and Effect of Scale

Experiment	Train Data	FID(Avg $\pm$ Std) $\downarrow$
Zero-shot	-	4.28 $\pm$ 0.60
1 $\times$ Augment	1xAug	2.44 $\pm$ 0.59
5 $\times$ Augment	5xAug	1.78 $\pm$ 0.24
10 $\times$ Augment	10xAug	1.69$\pm$0.08
50 $\times$ Augment	50xAug	2.30 $\pm$ 0.30
100 $\times$ Augment	100xAug	2.88 $\pm$ 1.05

5.2 Quality of Augmented Data

Augmented data achieved higher CLIP Score than seed data, representing an 11.54% improvement (Table 3). This suggests that the augmentation pipeline improves semantic alignment between image attributes and textual descriptions.

Table 3. Performance of Seed and Augmented Data

Train Data	CLIP(Avg $\pm$ Std) $\uparrow$
Seed	0.26 $\pm$ 0.20
1xAugment	0.29$\pm$0.13

5.3 Instruction and Image Following Performance

Two experiments were conducted using answer images and their corresponding instructions (sampling 100 pairs from augmented data). The first evaluated how well the original model and FIGMA aligned generated images with instructions

using CLIP Score, and the second measured how faithfully each model reconstructed the original images using LPIPS. FIGMA achieved a 19.23% higher CLIP Score and an 8.93% lower LPIPS than the base model, indicating better instruction adherence and improved preservation of visual details (Table 4).

Table 4. Following Performance of Each Model

Instruction Following Performance			
Experiment	Image Data	Text Data	CLIP(Avg±Std)↑
SD	Generated image	Answer instruction	0.26±0.08
FIGMA	Generated image	Answer instruction	0.31±0.04
Image Following Performance			
Experiment	Image Data	Image Data	LPIPS(Avg±Std)↓
SD	Generated image	Answer image	0.56±0.09
FIGMA	Generated image	Answer image	0.51±0.11

5.4 Ablation Study

To analyze the contribution of each component in the proposed pipeline, we conducted three experiments: (1) text augmentation only, (2) text augmentation combined with text expansion, and (3) the full pipeline. The full pipeline achieved the best FID of 1.98 (Table 5).

Table 5. Performance of Augmentation Strategies

Strategy	FID(Avg±Std)↓
Augment Text	2.81±0.19
Augment Text+Expand Text	2.21±0.09
Augment Text+Expand Text+Augment Image	1.98±0.11

To assess cross-domain generalization, FIGMA was evaluated on Nike, Zara, and FashionGen datasets. As shown in Table 6, FIGMA consistently outperformed the zero-shot baseline across all datasets. Furthermore, applying the augmented data to other diffusion models, including SD¹², SDXL¹³, and Kandinsky¹⁴¹⁵, also resulted in improved FID scores compared to their respective zero-shot baselines (Table 6). These results indicate that the proposed augmentation strategy generalizes beyond a single architecture.

¹² <https://huggingface.co/stablediffusionapi/deliberate-v2>

¹³ <https://huggingface.co/stabilityai/stable-diffusion-xl-base-1.0>

¹⁴ <https://huggingface.co/kandinsky-community/kandinsky-2-2-prior>

¹⁵ <https://huggingface.co/kandinsky-community/kandinsky-2-2-decoder>

Table 6. Generalization Performance

Model	Train Data	Zero-shot(FID↓)	Fine-tuned(FID↓)
SDXL	K-Fashion	1.93±0.06	1.87±0.05
Kandinsky	K-Fashion	10.21±0.68	10.00±0.53
FIGMA	Nike	11.86±0.22	10.36±0.12
FIGMA	Zara	5.97±0.12	5.91±0.07
FIGMA	FashionGen	43.67±0.13	37.96±0.05

6 Discussion

6.1 Limitations

Our model inherits biases from pretrained Stable Diffusion, limiting cultural diversity expression. First, as explained in Fig. 4, *Color* and *Print* attributes recorded the lowest match rates, suggesting difficulties in generating fine-grained visual textures and distinguishing subtle variations. In the second example, CLIP occasionally misclassifies fashion attributes. Fig. 5 and Table 7 show an error case where *suitable occasion*, *pattern* and *length* attributes were misclassified.

**Fig. 5.** Image of Error Case Example

6.2 Future Improvements

We propose: (1) Expanding training data with diverse cultural backgrounds and applying fairness-aware techniques, (2) Incorporating perceptual loss and additional regularization, and (3) Exploring HSV-based color correction and texture-aware loss functions for better Color and Print recognition [15].

7 Conclusion

We propose FIGMA, an instruction-based image generation model that combines automated data augmentation with domain-specific fine-tuning. Our method enables high-quality image generation with minimal manual annotation, addressing

Table 7. Correct and Misclassified Instruction of Error Case Example

[Correct Instruction] -Style: Feminine -Dress: Length: Maxi, Color: Beige, Collar: Round neck, Sleeve Length: Long Sleeve -Suitable occasion: Casual -Suitable season: Spring -Pattern: Solid
[Misclassified Instruction] -Style: Feminine -Dress: Length: Maxi, Color: Beige, Collar: Round neck, Sleeve Length: Short Sleeve -Suitable occasion: Formal Event -Suitable season: Spring -Pattern: Striped

key limitations of traditional data-centric approaches in terms of scalability, cost, and generalization.

Experimental results demonstrate that FIGMA significantly improves instruction-following accuracy and visual fidelity, as validated by FID, CLIP Score and LPIPS metrics. Moreover, our automated augmentation pipeline streamlines the data construction process, making it suitable for deployment in large-scale multimodal systems.

This work lays the foundation for a general-purpose, domain-agnostic instruction generation pipeline applicable to fields such as fashion, healthcare, industrial design, and beyond. We anticipate that FIGMA’s lightweight, modular architecture will enable future research in scalable, instruction-driven generative systems.

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